

Great question! You can **configure RAID on your Dell PowerEdge R640 entirely through iDRAC (Integrated Dell Remote Access Controller)** — **without needing a monitor, keyboard, or physical access** to the server.

This is one of the best features of Dell servers: **remote RAID setup, OS installation, and management** — all from your browser.

How to Create RAID via iDRAC (R640) — Step-by-Step

> This guide shows you how to:

- > - Access iDRAC
- > - Configure RAID (RAID 1, RAID 0, etc.)
- > - Install OS remotely using virtual media
- > - All without touching the server

Prerequisites

1. **PowerEdge R640** with iDRAC (Enterprise or Express)
2. **Network cable** connected from iDRAC port to your network
3. **iDRAC IP address** (assigned via DHCP or static)
4. **Admin credentials** (default: `root` / `calvin`)
5. **Proxmox, ESXi, or OS ISO file** on your PC
6. **Web browser** (Chrome, Firefox, Edge)

🌐 Step 1: Access iDRAC Web Interface

Option A: Find iDRAC IP

- Check your ****router/DHCP server**** for a device named `R640-iDRAC` or similar
- Or: Connect monitor/keyboard → reboot → watch iDRAC splash screen for IP

Option B: Set Static IP (Optional)

If no IP is assigned:

1. Connect keyboard + monitor
2. Reboot → Press ****F2**** → ****iDRAC Settings****
3. Set:
 - NIC Selection: ****Dedicated**** (or Shared)
 - IP Address: `192.168.1.101`
 - Netmask: `255.255.255.0`
 - Gateway: `192.168.1.1`
4. Save and exit

🔑 Step 2: Log in to iDRAC

Open a browser → go to:

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https://<iDRAC-IP>

...

Example: `https://192.168.1.101`

👉 Log in with:

- **Username**: `root`

- **Password**: `calvin` (default)

> 🔒 Change password after first login (recommended)

📁 Step 3: Launch Virtual Console (to Control Server Remotely)

In iDRAC web UI:

1. Go to **Overview → iDRAC Settings → Console** (or **Launch Console**)
2. Click **Launch Console** → Choose **HTML5** (no plugin needed)
3. Click **Connect** → You'll see the server's video output in your browser

✅ Now you can control the server as if you were sitting in front of it!

🛠️ Step 4: Enter RAID Configuration via iDRAC

Even better: You can configure RAID **without booting**, using **Storage → Virtual Disks**.

Method 1: Use iDRAC GUI (No Reboot Needed)

1. In iDRAC web UI → **Storage**
2. Click on your **Controller** (e.g., `PERC H740P`)
3. Scroll down → **Virtual Disk Management**
4. Click **Create** (or "Create Virtual Disk")

► Create RAID 1 for SSDs (Example)

Fill in:

- **RAID Level**: `RAID-1 (Mirror)`
- **Select Drives**: Choose your **2 SSDs**
- **Disk Size**: Max available
- **Name**: `SSD-Proxmox`
- **Read Policy**: `Read Ahead`
- **Write Policy**: `WriteBack` (if BBU is healthy)
- **Strip Size**: `64 KB`
- **Disk Cache Policy**: `Enabled`

 Click **Apply**

>  This will erase all data on the selected drives.

► Create RAID 1 for HDDs

Repeat:

- **RAID Level**: `RAID-1`
- **Select 2 HDDs**
- **Name**: `HDD-DATA-MIRROR`
- Same settings

☒ Click **Apply**

☒ Confirm Arrays Are Created

Go back to **Storage → Controller → Virtual Disks**

You should now see:

- `SSD-Proxmox` (RAID 1)
- `HDD-DATA-MIRROR` (RAID 1)

☒ RAID is now configured — **no need to press Ctrl+R during boot!**

🖥️ Step 5: Install OS Remotely (via Virtual Media)

Now install Proxmox or ESXi — **remotely**.

1. Mount ISO via iDRAC

In iDRAC web UI:

- Go to **Virtual Media**
- Click **Map CD/DVD**
- Click **Browse** → Select your **Proxmox or ESXi ISO**
- Click **Map Drive**

✅ The ISO is now "inserted" into the server virtually

2. Reboot Server to Install OS

In iDRAC:

- Go to **Power** → **Power Cycle** or **Graceful Restart**
- Launch **Virtual Console** again
- Watch boot process

It should boot from the **virtual CD/DVD (your ISO)**

3. Install OS Normally

Proceed with:

- Proxmox or ESXi installer
- Select the **SSD RAID array** (`SSD-Proxmox`) as target
- Complete installation

4. Unmount ISO After Install

After OS is installed:

- Go to **Virtual Media** → **Unmount** all ISOs
- Reboot

✅ Your server now runs the OS from the RAID array — all done remotely!

🔄 Optional: Dual-Boot via iDRAC (Switch OS Remotely)

If you're dual-booting **Proxmox + ESXi** (each on separate SSDs):

1. Use iDRAC to:

- **Power on** the server

- **Launch Console**

- Press **F11** → **Boot Manager**

2. Choose:

- `SSD-Proxmox` → Boots Proxmox

- `SSD-ESXi` → Boots ESXi

✅ You can switch hypervisors **remotely** at any time.

✅ Summary: iDRAC RAID Advantages

| Feature | Benefit |

|-----|-----|

| **No physical access needed** | Configure RAID from anywhere |

| **No monitor/keyboard** | Full remote setup |

| **GUI-based RAID** | Easier than Ctrl+R BIOS |

| **Virtual Media** | Mount ISOs remotely |

| **Console access** | See every boot screen |

📌 Tips

- 🗝️ Change default iDRAC password: **iDRAC Settings → Users**

- 📦 Enable **SSH** and **SNMP** for monitoring
- 🔄 Update firmware via **Maintenance → Firmware Update**
- 💡 Use **iDRAC9 Enterprise** for VLAN, dedicated IP, and better features

Need Help?

Let me know:

- Your iDRAC IP or login issues
- Which ISO you want to install
- If you want a **scripted or automated setup**

I'll guide you through every click! 💻 🔧